



```
In [1]: from pyspark.sql import SparkSession
```

```
In [2]: spark = SparkSession.builder \
        .master("local[*]") \
        .appName("ColabSpark") \
        .getOrCreate()
```

```
In [3]: data=spark.read.csv("a.csv.csv",header=True)
        data.show()
```

```

+---+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
| Id|          Name|Gender|Eyecolor|          Race|Haircolor|Height|
Publisher|Skincolor|Alignment|Weight|
+---+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
| 0|      A-Bomb| Male| yellow|          Human| No Hair| 203| M
arvel Comics| -| good| 441|
| 1|      Abe Sapien| Male| blue|      Ichthyo Sapien| No Hair| 191|Dark
Horse Comics| blue| good| 65|
| 2|      Abin Sur| Male| blue|          Ungaran| No Hair| 185|
DC Comics| red| good| 90|
| 3|      Abomination| Male| green|Human / Radiation| No Hair| 203| M
arvel Comics| -| bad| 441|
| 4|      Abraxas| Male| blue|      Cosmic Entity| Black| -99| M
arvel Comics| -| bad| -99|
| 5|      Absorbing Man| Male| blue|          Human| No Hair| 193| M
arvel Comics| -| bad| 122|
| 6|      Adam Monroe| Male| blue|          -| Blond| -99|
NBC - Heroes| -| good| -99|
| 7|      Adam Strange| Male| blue|          Human| Blond| 185|
DC Comics| -| good| 88|
| 8|      Agent 13|Female| blue|          -| Blond| 173| M
arvel Comics| -| good| 61|
| 9|      Agent Bob| Male| brown|          Human| Brown| 178| M
arvel Comics| -| good| 81|
| 10|      Agent Zero| Male| -|          -| -| 191| M
arvel Comics| -| good| 104|
| 11|      Air-Walker| Male| blue|          -| White| 188| M
arvel Comics| -| bad| 108|
| 12|      Ajax| Male| brown|          Cyborg| Black| 193| M
arvel Comics| -| bad| 90|
| 13|      Alan Scott| Male| blue|          -| Blond| 180|
DC Comics| -| good| 90|
| 14|      Alex Mercer| Male| -|          Human| -| -99|
Wildstorm| -| bad| -99|
| 15|      Alex Woolsly| Male| -|          -| -| -99|
NBC - Heroes| -| good| -99|
| 16|      Alfred Pennyworth| Male| blue|          Human| Black| 178|
DC Comics| -| good| 72|
| 17|      Alien| Male| -|      Xenomorph XX121| No Hair| 244|Dark
Horse Comics| black| bad| 169|
| 18|      Allan Quatermain| Male| -|          -| -| -99|
Wildstorm| -| good| -99|
| 19|      Amazo| Male| red|          Android| -| 257|
DC Comics| -| bad| 173|
+---+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
only showing top 20 rows

```

```
In [4]: data.orderBy("Height").show()
```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+								
+-----+-----+-----+-----+-----+-----+-----+-----+-----+								
Id	Name	Gender	Eyecolor	Race	Haircolor	Height	Publis	
her	Skincolor	Alignment	Weight					
+-----+-----+-----+-----+-----+-----+-----+-----+-----+								
+-----+-----+-----+-----+-----+-----+-----+-----+-----+								
76	Beetle	Male	-	-	-	-99	Marvel	Com
ics	-	bad	-99					
125	Blue Beetle III	Male	brown	Human	Black	-99	DC	Com
ics	-	good	-99					
77	Ben 10	Male	-	-	-	-99	DC	Com
ics	-	good	-99					
4	Abraxas	Male	blue	Cosmic Entity	Black	-99	Marvel	Com
ics	-	bad	-99					
79	Beyonder	Male	-	God / Eternal	-	-99	Marvel	Com
ics	-	good	-99					
14	Alex Mercer	Male	-	Human	-	-99	Wildst	
orm	-	bad	-99					
81	Big Daddy	Male	-	-	-	-99	Icon	Com
ics	-	good	-99					
18	Allan Quatermain	Male	-	-	-	-99	Wildst	
orm	-	good	-99					
83	Bill Harken	Male	-	Alpha	-	-99	S	
yFy	-	good	-99					
26	Angela	Female	-	-	-	-99	Image	Com
ics	-	bad	-99					
84	Billy Kincaid	Male	-	-	-	-99	Image	Com
ics	-	bad	-99					
35	Aquababy	Male	blue	-	Blond	-99	DC	Com
ics	-	good	-99					
86	Bionic Woman	Female	blue	Cyborg	Black	-99	N	
ULL	-	good	-99					
45	Arsenal	Male	-	Human	-	-99	DC	Com
ics	-	good	-99					
87	Bird-Brain	-	-	-	-	-99	Marvel	Com
ics	-	good	-99					
50	Atom	Male	-	-	-	-99	DC	Com
ics	-	good	-99					
88	Bird-Man	Male	-	Human	-	-99	Marvel	Com
ics	-	bad	-99					
54	Atom IV	Male	brown	-	Black	-99	DC	Com
ics	-	good	72					
62	Batgirl	Female	-	-	-	-99	DC	Com
ics	-	good	-99					
64	Batgirl III	Female	-	-	-	-99	DC	Com
ics	-	good	-99					

only showing top 20 rows

Find the count of male heroes.

```
In [9]: data.filter(data.Gender=="Male").count()
```

Out[9]: 505

4. Find the count of Female heroes.

```
In [10]: data.filter(data.Gender=="Female").count()
```

Out[10]: 200

5. Find the count of the particular race.

```
In [12]: data.groupby("Race").count().show()
```

```
+-----+-----+
|          Race|count|
+-----+-----+
|      Neyaphem|    1|
|      Symbiote|    9|
|    Kryptonian|    7|
|      Ungaran|    1|
|      Martian|    1|
|    Strontian|    1|
| Human / Radiation|  11|
| Kakarantharaian|    1|
|    Metahuman|    2|
|Dathomirian Zabrak|    1|
|  Human / Cosmic|    2|
| Xenomorph XX121|    1|
|        Animal|    4|
|    Frost Giant|    2|
|        Cyborg|   11|
|      Inhuman|    4|
|   God / Eternal|   14|
| Yoda's species|    1|
|   Ichthyo Sapien|    1|
|        Rodian|    1|
+-----+-----+
```

only showing top 20 rows

6. Find the average height of characters in each publisher

```
In [46]: from pyspark.sql.functions import col
```

```
data2 = data.withColumn("height1",col("Height").cast("double"))
```

```
In [47]: data2.filter(col("height1").isNotNull()).groupBy("Publisher").avg("height1").s
```

Publisher	avg(height1)
Rebellion	188.0
HarperCollins	-99.0
J. R. R. Tolkien	279.0
Star Trek	-5.5
Marvel Comics	142.75644329896906
Wildstorm	-99.0
NULL	-2.3666666666666667
South Park	-99.0
Sony Pictures	179.0
Titan Books	183.0
ABC Studios	-99.0
SyFy	-99.0
Image Comics	-76.85714285714286
Universal Studios	-99.0
IDW Publishing	-99.0
NBC - Heroes	-99.0
Hanna-Barbera	-99.0
DC Comics	91.07209302325582
Microsoft	213.0
J. K. Rowling	-99.0

only showing top 20 rows

```
In [ ]: data.groupby(" Publisher").avg("height").show()
```

7. Write a sql query to select superheroes with blue eyes

```
In [51]: data.createOrReplaceTempView("EMP")
spark.sql("SELECT *FROM EMP WHERE Eyecolor = 'blue'").show()
```

```

+---+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
| Id|          Name|Gender|Eyecolor|          Race|Haircolor|Height|          P
ublisher|Skincolor|Alignment|Weight|
+---+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
| 1|      Abe Sapien|  Male|    blue|Ichthyo Sapien|  No Hair|   191|Dark Hors
e Comics|    blue|    good|   65|
| 2|      Abin Sur|  Male|    blue|    Ungaran|  No Hair|   185|      D
C Comics|    red|    good|   90|
| 4|      Abraxas|  Male|    blue|Cosmic Entity|   Black|  -99|    Marve
l Comics|    -|    bad|  -99|
| 5|  Absorbing Man|  Male|    blue|    Human|  No Hair|   193|    Marve
l Comics|    -|    bad|  122|
| 6|    Adam Monroe|  Male|    blue|    -|   Blond|  -99|    NBC
- Heroes|    -|    good|  -99|
| 7|    Adam Strange|  Male|    blue|    Human|   Blond|   185|      D
C Comics|    -|    good|   88|
| 8|      Agent 13|Female|    blue|    -|   Blond|   173|    Marve
l Comics|    -|    good|   61|
| 11|   Air-Walker|  Male|    blue|    -|   White|   188|    Marve
l Comics|    -|    bad|  108|
| 13|    Alan Scott|  Male|    blue|    -|   Blond|   180|      D
C Comics|    -|    good|   90|
| 16|Alfred Pennyworth|  Male|    blue|    Human|   Black|   178|      D
C Comics|    -|    good|   72|
| 22|      Angel|  Male|    blue|    -|   Blond|   183|    Marve
l Comics|    -|    good|   68|
| 27|   Animal Man|  Male|    blue|    Human|   Blond|   183|      D
C Comics|    -|    good|   83|
| 29|    Ant-Man|  Male|    blue|    Human|   Blond|   211|    Marve
l Comics|    -|    good|  122|
| 30|   Ant-Man II|  Male|    blue|    Human|   Blond|   183|    Marve
l Comics|    -|    good|   86|
| 33|   Anti-Venom|  Male|    blue|  Symbiote|   Blond|   229|    Marve
l Comics|    -|    -|  358|
| 35|   Aquababy|  Male|    blue|    -|   Blond|  -99|      D
C Comics|    -|    good|  -99|
| 36|   Aqualad|  Male|    blue|  Atlantean|   Black|   178|      D
C Comics|    -|    good|  106|
| 37|   Aquaman|  Male|    blue|  Atlantean|   Blond|   185|      D
C Comics|    -|    good|  146|
| 38|   Arachne|Female|    blue|    Human|   Blond|   175|    Marve
l Comics|    -|    good|   63|
| 39|   Archangel|  Male|    blue|    Mutant|   Blond|   183|    Marve
l Comics|   blue|    good|   68|
+---+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
only showing top 20 rows

```

In []:

In []: